# MANZIL

FOR JEE ASPIRANTS

ONESHOT

# PROBABILITY

SACHIN JAKHAR

# Topics To Be Covered

1. General Terminology, Definition
2. Probability using P&C & Set Theory
3. Independent Events & Conditional Probability
4. Total Probability Theorem & Baye's Theorem
5. Probability Distribution, Random Variable & B.P.D. and PYQ's

# GENERAL TERMINOLOGY

1. ***Random Experiment :*** Experiment whose outcome is known.

   (Throwing a dice, tossing a coin, drawing a card.....)

2. ***Sample Space :*** Set of all possible outcomes. ✓

3. ***Event :*** Subset of sample space ✓

   ***Simple Event :*** Singleton set wali event

   ***Mixed or Compound Events :*** Events having more than one outcome.

Ques :-

Experiment : Two coins tossed.

$S = \{(H,H)\ (H,T)\ (T,H)\ (T,T)\}$

Sample Space.

Event (E) = getting atleast 1 Head

Mixed

$= \{(H,H)\ (H,T)\ (T,H)\}$

Subset.

# GENERAL TERMINOLOGY

4. *Types of Events :*

(i) **Equally Likely Events :** Having equal chances of occurrence.

(ii) **Mutually Exclusive/Disjoint/Incompatible Events:**

Events whose simultaneous occurrence is impossible.

$E_1 \cap E_2 = \phi$

(iii) **Mutually Exhaustive Events:**

Events whose union is sample space.

$E_1 \cup E_2 = S$

mut. exht.

# CLASSICAL (a-priori) DEFINITION

If an random experiment has total $(m+n)$ mutually exclusive, mutually exhaustive & equally likely outcomes, out of which m favours event 'E' then probability of occurrence of Event E is denoted by P(E)

$$P(E) = \frac{\text{no. of fav. outcomes}}{\text{Total outcomes}}$$

$S = \{O_1, O_2, O_3, O_4, O_5\}$

$E = \{O_2, O_3, O_5\}$

$P(E) = \frac{3}{5}$

## Important Note

1. Sum of probabilities of each sample point or probability of sample space = 1
   If sample space (S) = $\{x_1, x_2, x_3, \ldots, x_n\} \Rightarrow \sum_{i=1}^{n} P(x_i) = 1$
2. If $P(E) = 0$, then event E is Impossible event.
3. If $P(E) = 1$, then event E is Sure event.

$$0 \leq P(E) \leq 1$$

4. Odds in favour of any event E = $\left(\frac{\text{Favourable}}{\text{unfavourable}}\right)$ ✓
5. Odds against event E = $\left(\frac{\text{unfavourable}}{\text{favourable}}\right)$ ✓
6. Probability of non-occurrence of event E = $P(\bar{E}) = 1 - P(E)$

bar/complement.

Ques:

$n(S) = 11$

→ fav. to E = 8

→ unfav. to E = 3.

$P(E) = \frac{8}{11}$.

Odd in fav. $= \frac{\text{fav.}}{\text{unfav.}} = \frac{8}{3}$.

Odds against $= \frac{\text{unfav.}}{\text{fav.}} = \frac{3}{8}$.

Total = 7 → F → 3, UF → 4

ex:- $P(E) = \frac{3}{7}$

Odds against E $= \frac{4}{3}$.

**NOTE**

If all outcomes of sample space are not equally likely.
P(E) = Sum of probabilities of all the favourable outcomes of E.

$S = \{O_1, O_2, O_3, O_4, O_5\}$ — $p$ $p$ $p$ $p$ $p$

net prob. kaa sum = 1

$5p = 1$

$p = 1/5$.

formula applicable ✓

$E_1 = \{O_1, O_3, O_4\}$

$P(E_1) = \frac{3}{5}$

$P(E_1) = P(O_1) + P(O_3) + P(O_4)$

$= \frac{1}{5} + \frac{1}{5} + \frac{1}{5} = \frac{3}{5}$

$S = \{O_1, O_2, O_3, O_4, O_5\}$ — $\frac{3}{10}$ $\frac{1}{10}$ $\frac{3}{10}$ $\frac{1}{10}$ $\frac{2}{10}$.

formula applicable ✗

$E_1 = \{O_1, O_3, O_4\}$

$P(E_1) = \frac{3}{5}$

✗

$P(E_1) = P(O_1) + P(O_3) + P(O_4)$

$= \frac{3}{10} + \frac{3}{10} + \frac{1}{10}$

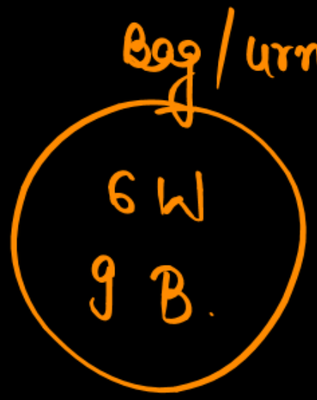
$= \frac{7}{10}$ ✓

## QUESTION [JEE Mains-2024]

[Ans. : (C)]

An urn contains 6 white and 9 black balls. Two successive draws of 4 balls are made without replacement. The probability, that the first draw gives all white balls and the second draw gives all black balls, is

1. $\frac{5}{256}$
2. $\frac{5}{715}$
3. $\frac{3}{715}$
4. $\frac{3}{256}$




$S = \{ \ldots\ldots\ldots\ldots \}$

Total outcomes $= {}^{15}C_4 \times {}^{11}C_4$

$E = \{ \quad \}$

fav. $= {}^{6}C_4 \times {}^{9}C_4$

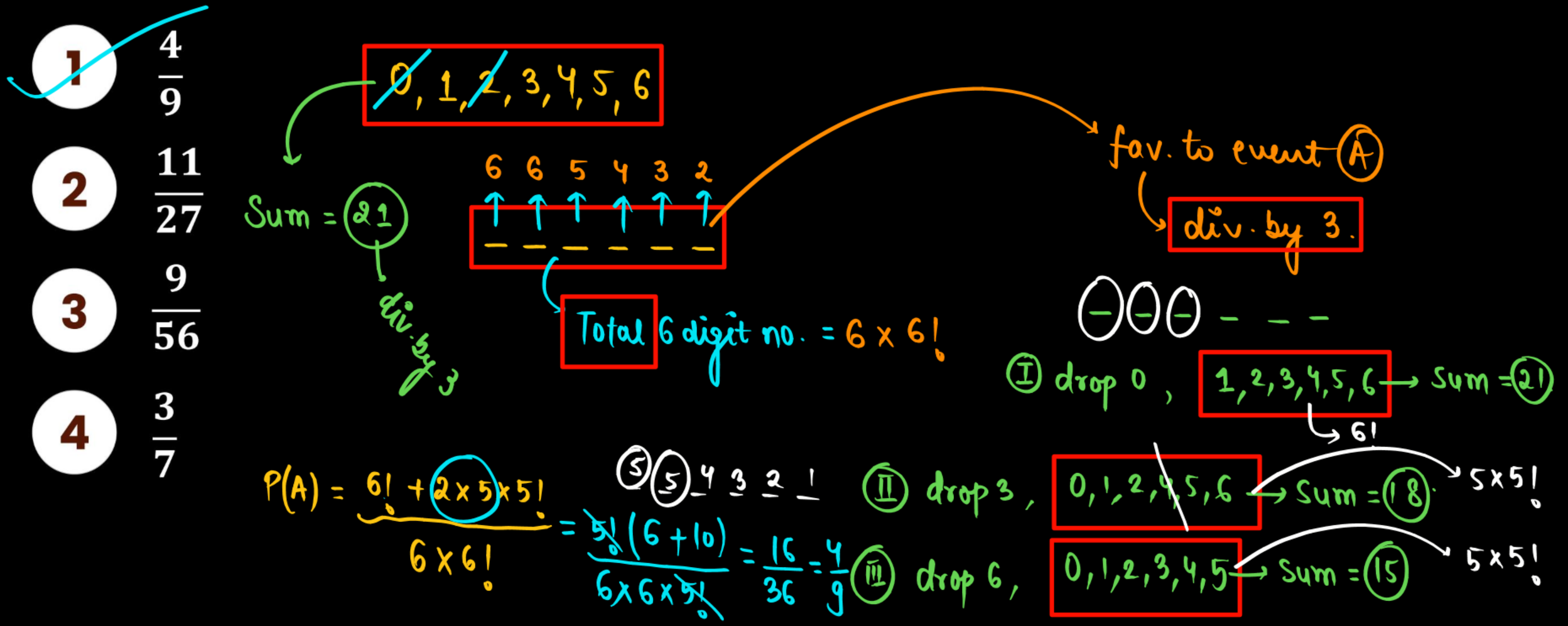
$$P(E) = \frac{{}^{6}C_4 \times {}^{9}C_4}{{}^{15}C_4 \times {}^{11}C_4} = \frac{3}{715}$$

## QUESTION [JEE Mains-2021]

[Ans. : (A)]

Let $A$ denote the event that a 6-digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, is divisible by 3. Then probability of event $A$ is equal to

1. $\frac{4}{9}$
2. $\frac{11}{27}$
3. $\frac{9}{56}$
4. $\frac{3}{7}$



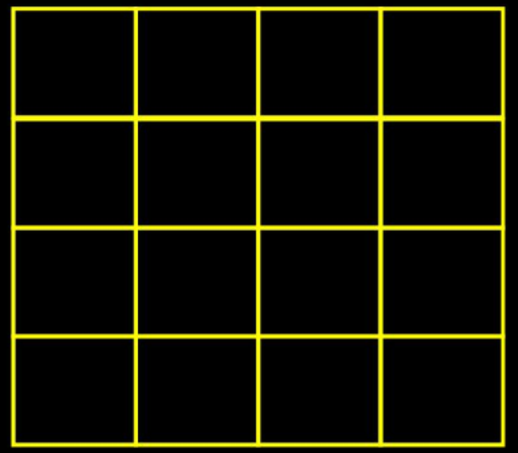
**QUESTION [JEE Mains-2025]**

[Ans. : (A)]

A board has 16 squares as shown in the figure

Out of these 16 squares, two squares are chosen at random. The probability that they have no side in common is:

1. $\frac{4}{5}$
2. $\frac{3}{5}$
3. $\frac{7}{10}$
4. $\frac{23}{30}$



\# Total $= {}^{16}C_2$

- 1 side common $= 3\times4 + 3\times4 = 24$
- No side common

Fav. = no side common $= {}^{16}C_2 - 24$.

$$\text{Prob} = \frac{{}^{16}C_2 - 24}{{}^{16}C_2} = 1 - \frac{24}{120} = 1 - \frac{1}{5} = \frac{4}{5}$$

## QUESTION [JEE Mains-2025]

[Ans. : (B)]

Let $A = [a_{ij}]$ be a square matrix of order 2 with entries either 0 or 1. Let $E$ be the event that $A$ is an invertible matrix. Then the probability $P(E)$ is

1. $\frac{5}{8}$
2. $\frac{3}{8}$
3. $\frac{3}{16}$
4. $\frac{1}{8}$

$\{0,1\}$

$A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}_{2\times 2}$

Total $= 2^4 = 16$

$P(E) = \frac{6}{16} = \frac{3}{8}$

Fav. → Invertible ($|A| \neq 0$)

1111 → $\phi$.

1110

$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

④

1100

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

2

1000 → $\phi$

0000 → $\phi$.

## QUESTION [JEE Mains-2025]

[Ans. : (D)]

Let $S$ be the set of all the words that can be formed by arranging all the letters of the word GARDEN. From the set $S$, one word is selected at random. The probability that the selected word will NOT have vowels in alphabetical order is

1. $\frac{1}{3}$
2. $\frac{2}{3}$
3. $\frac{1}{4}$
4. $\frac{1}{2}$

# Total words = $\frac{6!}{2}$

GARDEN

A, E — G, R, D, N

A पहले E बादमें → Alph order.

E पहले A बादमें → NOT Alph ord.

ARDEGN

– E – – A –

${}^6C_2 \times 1 \times 4!$

$= 15 \times 24$

$= 360$

## QUESTION [JEE Mains-2025]

All five letter words are made using all the letters $A, B, C, D, E$ and arranged as in an English dictionary with serial numbers. Let the word at serial number $n$ be denoted by $W_n$. Let the probability $P(W_n)$ of choosing the word $W_n$ satisfy $P(W_n) = 2P(W_{n-1}), n > 1$. If $P(CDBEA) = \frac{2^{\alpha}}{2^{\beta}-1}, \alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to

$A, B, C, D, E$

Total words $= 5! = 120$

$n=2$ $P(W_2) = 2P(W_1) = 2^1 p$

$n=3$ $P(W_3) = 2P(W_2) = 2^2 P(W_1) = 2^2 p$

$P(W_4) = 2P(W_3) = 2^3 P(W_1) = 2^3 p$

$p\{2^0 + 2^1 + \ldots + 2^{119}\} = 1$

$p \times 1\left(\frac{2^{120}-1}{2-1}\right) = 1$

$p = \frac{1}{2^{120}-1}$

Rank of word in Dict.

1 → $W_1$

2 → $W_2$

3 → $W_3$

120 → $W_{120}$

$S = \{W_1, W_2, W_3, \ldots, W_{120}\}$

$p + 2p + 2^2p + 2^3p \ldots + 2^{119}p = 1$

## QUESTION [JEE Mains-2025]

$\begin{cases} \alpha = 63 \\ \beta = 120 \end{cases}$ fnf.

183

[Ans. : 183]

All five letter words are made using all the letters $A, B, C, D, E$ and arranged as in an English dictionary with serial numbers. Let the word at serial number $n$ be denoted by $W_n$. Let the probability $P(W_n)$ of choosing the word $W_n$ satisfy $P(W_n) = 2P(W_{n-1}), n > 1$. If $P(CDBEA) = \frac{2^{\alpha}}{2^{\beta}-1}, \alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to

A, B, C, D, E

Total words = $5! = 120$.

Rank of word in Dict.

1 → $W_1$

2 → $W_2$

3 → $W_3$

64 CDBEA → $W_{64}$

120 → $W_{120}$

Rank or S.No. of CDBEA

A - - - - → $4!$

B - - - - → $4!$

CA - - - → $3!$

CB - - - → $3!$

CDA - - → $2!$

CDBAE → 1

CDBEA → 1

64

$p\{2^0 + 2^1 + \ldots + 2^{119}\} = 1$

$p \times 1\left(\frac{2^{120}-1}{2-1}\right) = 1$

$p = \frac{1}{2^{120}-1}$.

$P(W_{64}) = 2^{63} \times p$

$= \frac{2^{63}}{2^{120}-1}$

## QUESTION [JEE Mains-2025]

$S = \{4\lambda+1, 4\lambda+2, 4\lambda+3, 4\lambda\}$

Two number $k_1$ and $k_2$ are randomly chosen from the set of natural numbers. Then, the probability that the value of $i^{k_1} + i^{k_2}, (i = \sqrt{-1})$ is non-zero, equals

1. $\frac{2}{3}$
2. $\frac{1}{2}$
3. $\frac{1}{4}$
4. $\frac{3}{4}$

$i = \sqrt{-1} = i^5 = i^9 \ldots = i^{4\lambda+1}$

$i^2 = -1 = i^6 = i^{10} \ldots = i^{4\lambda+2}$

$i^3 = -i = i^7 = i^{11} \ldots = i^{4\lambda+3}$

$i^4 = 1 = i^8 = i^{12} \ldots = i^{4\lambda}$

$i^{k_1} + i^{k_2} =$ zero (Not Req.)

$i^{k_1}$

$\begin{cases} i^{4\lambda+1} + i^{4\lambda+3} = i + (-i) = 0 \\ i^{4\lambda+3} + i^{4\lambda+1} = -i + i = 0 \end{cases}$

$\begin{cases} i^{4\lambda+2} + i^{4\lambda} = -1 + 1 = 0 \\ i^{4\lambda} + i^{4\lambda+2} = 1 + (-1) = 0 \end{cases}$

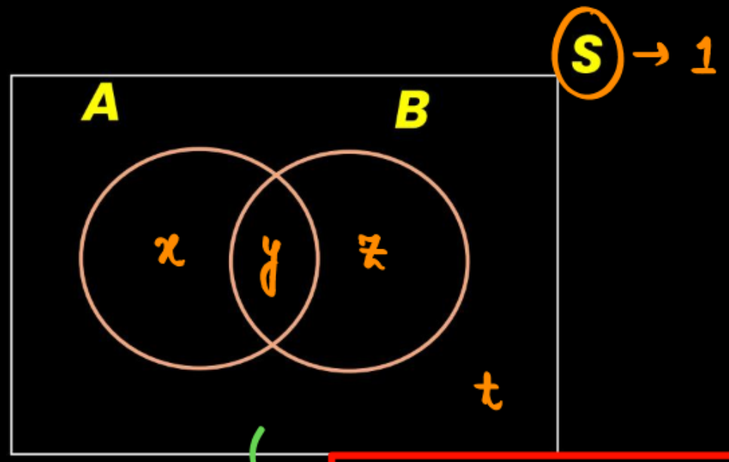
$P(\text{sum zero}) = \frac{1}{4}^{k_1} \times \frac{1}{4}^{k_2} \times 2 + 2 \times \frac{1}{4}^{k_1} \times \frac{1}{4}^{k_2}$

$= \frac{4}{16} = \frac{1}{4}$

$P(\overline{\text{sum zero}}) = 1 - \frac{1}{4} = \frac{3}{4}$

# USE OF SET THEORY

## # For TWO SETS:




$x+y+z+t=1$.

**Probability of A & B =**

**Probability of A or B =**

**If A & B are mutually exclusive**

**Probability of non-occurrence of A**

$n(A\cup B) = n(A)+n(B)-n(A\cap B)$

$\div n(S)$

$$P(A\cup B) = P(A)+P(B)-P(A\cap B)$$

$P(A) = x+y$

$P(B) = y+z$.

$P(A\cap B) = y$

$P(A\cup B) = x+y+z$.

$P(\bar{A}) = z+t$

$= 1-(x+y)$

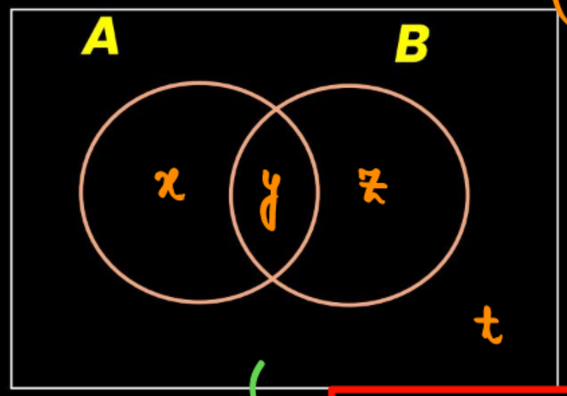
$P(\bar{B}) = x+t = 1-(y+z)$

$P(A\cap\bar{B}) = x$

↳ exactly A

# USE OF SET THEORY

## # For TWO SETS:




$x+y+z+t = 1$

**Probability of A & B =** $P(A \cap B) = P(AB)$

**Probability of A or B =** $P(A \cup B) = P(A+B)$

**If A & B are mutually exclusive** $\Rightarrow P(A \cap B) = y = 0$

**Probability of non-occurrence of A** $= 1 - P(A)$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$\div n(S)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = t$

$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = x+z+t$

$P(\bar{A} \cup B) = t+z+y$

$P(\bar{A} \cap B) = z$

De-Morgan's law

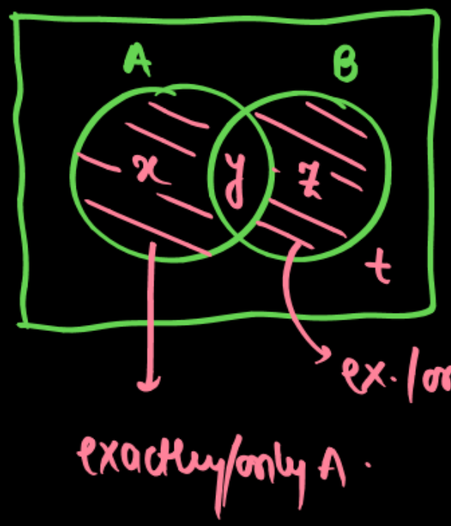
$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

## QUESTION [JEE Mains-2020]

Let $A$ and $B$ be two events such that the probability that exactly one of them occurs is $\frac{2}{5}$ and the probability that $A$ or $B$ occurs is $\frac{1}{2}$, then the probability of both of them occur together is

1. 0.02
2. 0.10
3. 0.01
4. 0.20




$A \cup B$

$S \rightarrow 1$. $\quad \cap \rightarrow y = ?$

$x + y + z + t = 1$.

exactly one $= x + z = \frac{2}{5}$.

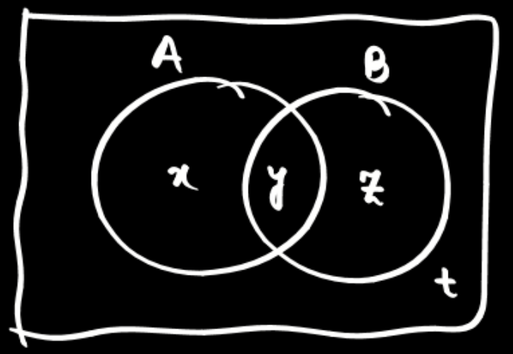
$x + y + z = \frac{1}{2}$

$y = \frac{1}{2} - \frac{2}{5} = \frac{5-4}{10} = \frac{1}{10} = 0.1$

## QUESTION [IIT-JEE-1987]

The probability that at least one of the events $A$ and $B$ occurs is 0.6. If $A$ and $B$ occur simultaneously with probability 0.2, then $P(\bar{A}) + P(\bar{B})$ is equal to

1. 0.4
2. 0.8
3. 1.2
4. 1.4




$A \cup B$

$A \cap B$

$S \to 1$

$x + y + z + t = 1$

$x + y + z = \frac{6}{10}$

$y = \frac{2}{10}$

$t = 1 - \frac{6}{10}$

$t = \frac{4}{10}$

$x + z = \frac{6}{10} - \frac{2}{10}$

$= \frac{4}{10}$

$P(\bar{A}) + P(\bar{B})$

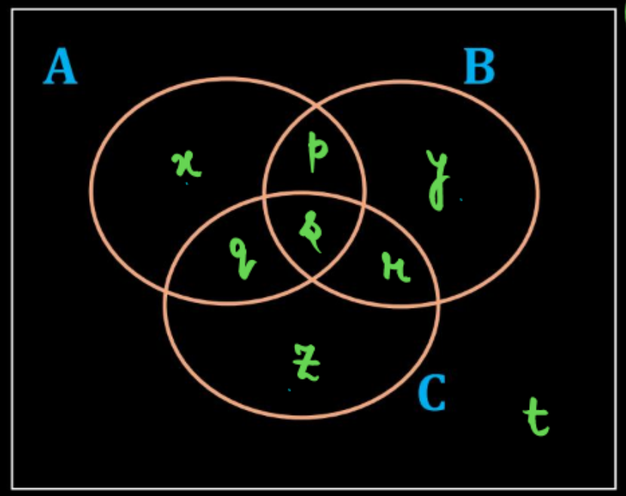
$(z + t) + (x + t)$

$(x + z + 2t) = 1.2$

$\frac{4}{10} + 2 \times \frac{4}{10} = \frac{12}{10}$

# USE OF SET THEORY

## # For THREE SETS :




$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$\bar{A} \cap \bar{B} \cap C = z$.

$A \cap B \cap \bar{C} = p$.

exactly 1 $= x + y + z$.

exactly 2 $= p + q + r$.

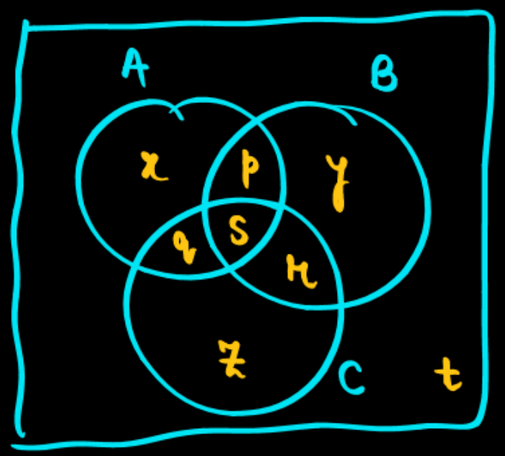
atleast 1 $= A \cup B \cup C$
$= x + y + z + p + q + r + s$.

atleast 2 $=$
$= p + q + r + s$.

## QUESTION [IIT-JEE-2002]

If $P(B) = \frac{3}{4}, P(A \cap B \cap \bar{C}) = \frac{1}{3}$ and $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, then $P(B \cap C)$ is equal to

1. $\frac{1}{12}$
2. $\frac{1}{6}$
3. $\frac{1}{15}$
4. $\frac{1}{9}$




$S \rightarrow (1)$

$p + (s+r) + y = \frac{3}{4}$

$p = \frac{1}{3}$

$y = \frac{1}{3}$

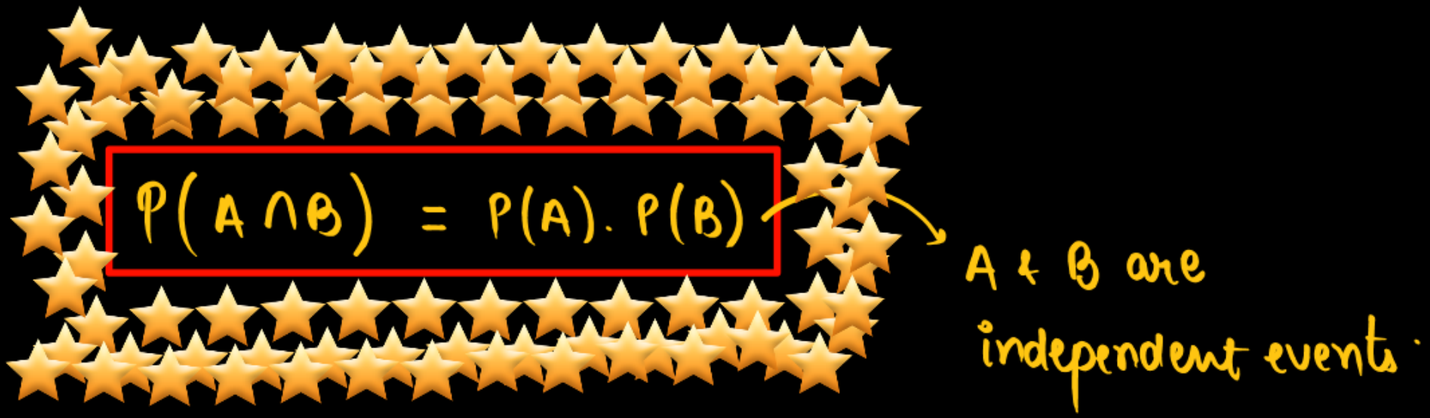
$s + r = \frac{3}{4} - \frac{1}{3} - \frac{1}{3}$

$= \frac{3}{4} - \frac{2}{3}$

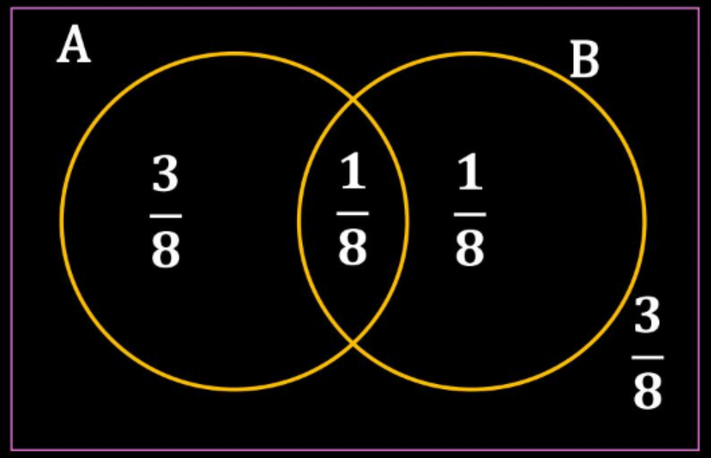
$= \frac{1}{12}$

# INDEPENDENT EVENTS

**Independent events-Events $A$ and $B$ are said to be independent if occurrences or non-occurrence of one doe not affect the probability of occurrence or non-occurrence of the other.**



# EXAMPLE




S → (1)

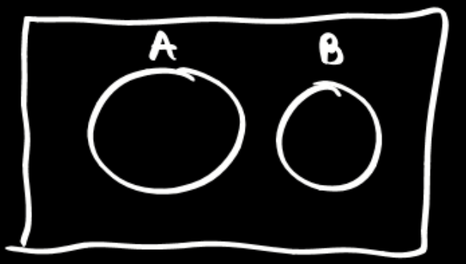
$$P(A) = \frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

$$P(B) = \frac{1}{8} + \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$$

$$P(A \cap B) = \frac{1}{8}$$

A & B are indep. events.

Mutually exclusive




# $P(A \cap B) = 0$

**NOTE**

1. If A & B are independent events then all possible combinations like $(\overline{A}, B)$ or $(A, \overline{B})$ or $(\overline{A}, \overline{B})$ are also independent events.
2. Independent events & mutually exclusive events are different things.
3. If A, B & C are three independent events then :

$$P(A \cap B) = P(A) \cdot P(B)$$
$$P(B \cap C) = P(B) \cdot P(C)$$
$$P(A \cap C) = P(A) \cdot P(C)$$

} pair wise indep.

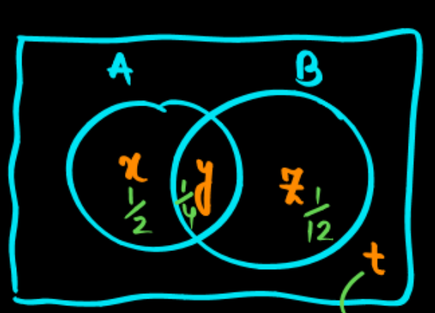
$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

A, B & C are indep. events.

## QUESTION [JEE Mains-2014]

Let $A$ and $B$ be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$, where $\overline{A}$ stands for the complement of the event $A$. Then the events $A$ and $B$ are

1. Independent but not equally likely
2. Independent and equally likely
3. Mutually exclusive and independent
4. Equally likely but not independent




$S \rightarrow 1$.

$P(A) = x + y = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

$P(B) = y + z = \frac{1}{4} + \frac{1}{12}$

$= \frac{3+1}{12} = \frac{1}{3}$ ✓

$P(A \cap B) = \frac{1}{4}$

$t = \frac{1}{6}$

$y = \frac{1}{4}$

$z + t = \frac{1}{4} \Rightarrow z = \frac{1}{4} - \frac{1}{6} = \frac{2}{24} = \frac{1}{12}$

$x + y + z + t = 1$

## QUESTION [IIT-JEE-2008]

An experiment has 10 equally likely outcomes. Let $A$ and $B$ be two non - empty events of the experiment. If $A$ consists of 4 outcomes, then the number of outcomes that $B$ must have, so that $A$ and $B$ are independent, is

1. 2, 4 or 8
2. 3, 6 or 9
3. 4 or 8
4. 5 or 10

$S = \{O_1, O_2, O_3 \ldots O_{10}\}$ (each $\frac{1}{10}$)

$A = \{-, -, -, -\}$ → $P(A) = \frac{2}{5}$

$B = \{-, -, \ldots\ldots -\}$ → Outcomes = $n$ → $P(B) = \frac{n}{10}$

$A \cap B = \{- - \ldots\ldots\}$ → Outcomes = 0, 1, 2, 3, 4

$P(A \cap B) = P(A) \cdot P(B)$

$\frac{0}{10}, \frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10} = \frac{2}{5} \times \frac{n}{10}$

$\frac{0}{10}, \frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10} = \frac{n}{25}$

$n = 0$ ✗ $\quad n = \frac{25}{10}$, $n = \frac{5}{2}$ ✗ $\quad n = \frac{50}{10}$, $n = 5$ ✓ $\quad n = \frac{75}{10}$, $n = \frac{25}{2}$ ✗ $\quad n = \frac{100}{10}$, $n = 10$ ✓

# CONDITIONAL PROBABILITY

$P(E_1) = \frac{3}{4}$ $\quad P(E_2) = \frac{3}{4}$.

ex: Exp: Two coins are tossed.

$S = \{(H,H)\ (H,T)\ (T,H)\ (T,T)\}$

$E_1$ = getting atleast 1 Head. $= \{(HH)\ (HT)\ (TH)\}$

$E_2$ = getting atleast 1 tail. $= \{(HT)\ (TH)\ (TT)\}$

$E_1 \cap E_2 = \{(HT)\ (TH)\}$

# $P(E_1/E_2)$ = P(event $E_1$ if $E_2$ is already occured)

$E_2$ हो चुकी है

↓

New sample space.

$$= \frac{n(E_1 \cap E_2)}{n(E_2)} = \frac{P(E_1 \cap E_2)}{P(E_2)}$$

$$= \frac{2/4}{(3/4)} = \frac{2}{3}$$

# CONDITIONAL PROBABILITY

$E_1$ = getting atleast 1 Head = $\{(HT)(TH)\}$

$E_2$ = getting atleast 1 tail. = $\{(HT)(TH)(TT)\}$

New sample space

# $P(E_1/E_2)$ = P(event $E_1$ if $E_2$ is already occured)

= हो चुकी ई' पे = $\frac{2}{3}$

Ques:

Two dice are rolled together.

* $E_1$ = sum of outcomes is 7. = $\{(1,6)\ (6,1)\ (2,5)\ (5,2)\ (3,4)\ (4,3)\}$

$E_2$ = exactly (1) outcome is even. = $\{(1,6)\ (6,1)\ (2,5)\ (5,2)\ (3,4)\ (4,3)\}$

$$P\left(\frac{E_2}{E_1}\right) = \frac{6}{6} = 1$$

$E_1$ → new sample space.

Ques:

Two dice are rolled together.

* $E_1$ = sum of outcomes is 8 = $\{(2,6)\ (6,2)\ (5,3)\ (3,5)\ (4,4)\}$

$E_2$ = atleast (1) outcome is even. = $\{(2,6)\ (6,2)\ (4,4)\}$

$$P\left(\frac{E_2}{E_1}\right) = \frac{3}{5} = \frac{P(E_1 \cap E_2)}{P(E_1)}$$

→ new sample space.

# CONDITIONAL PROBABILITY

$P\left(\frac{A}{B}\right)$ = Probability of occurrence of event $A$ when event $B$ already occurred

$$P\left(\frac{A}{B}\right) = \frac{n(A \cap B)}{n(B)} = \frac{P(A \cap B)}{P(B)}$$

$P\left(\frac{B}{A}\right)$ = Probability of occurrence of event $B$ when event $A$ already occurred

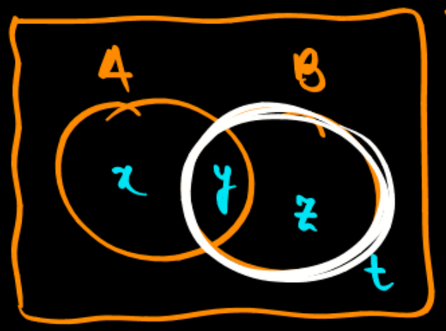
$$P\left(\frac{B}{A}\right) = \frac{n(A \cap B)}{n(A)} = \frac{P(A \cap B)}{P(A)}$$

## QUESTION [JEE Mains-2025] C.P. + S.T.

If $A$ and $B$ are two events such that $P(A \cap B) = 0.1$, and $P(A|B)$ and $P(B|A)$ are the roots of the equation $12x^2 - 7x + 1 = 0$, then the value of $\frac{P(\bar{A} \cup \bar{B})}{P(\bar{A} \cap \bar{B})}$ is

1. $\frac{5}{3}$
2. $\frac{4}{3}$
3. $\frac{7}{4}$
4. $\frac{9}{4}$

$12x^2 - 7x + 1 = 0 \Rightarrow x = \frac{1}{3}, \frac{1}{4}$




$S \to 1$.

$P(A/B) = \frac{1}{3} = \frac{y}{y+z} \Rightarrow y + z = 3y$

$z = 2y$

$P(B/A) = \frac{1}{4} = \frac{y}{x+y}$

$x + y = 4y$

$x = 3y$

$y = \frac{1}{10}$

$x + y + z + t = 1$.

$3y + y + 2y + t = 1$

$t = 1 - 6y = 1 - \frac{6}{10} = \frac{4}{10}$

$P(\bar{A} \cup \bar{B}) = x + z + t$

$= 3y + 2y + \frac{4}{10}$

$= \frac{5}{10} + \frac{4}{10} = \frac{9}{10}$.

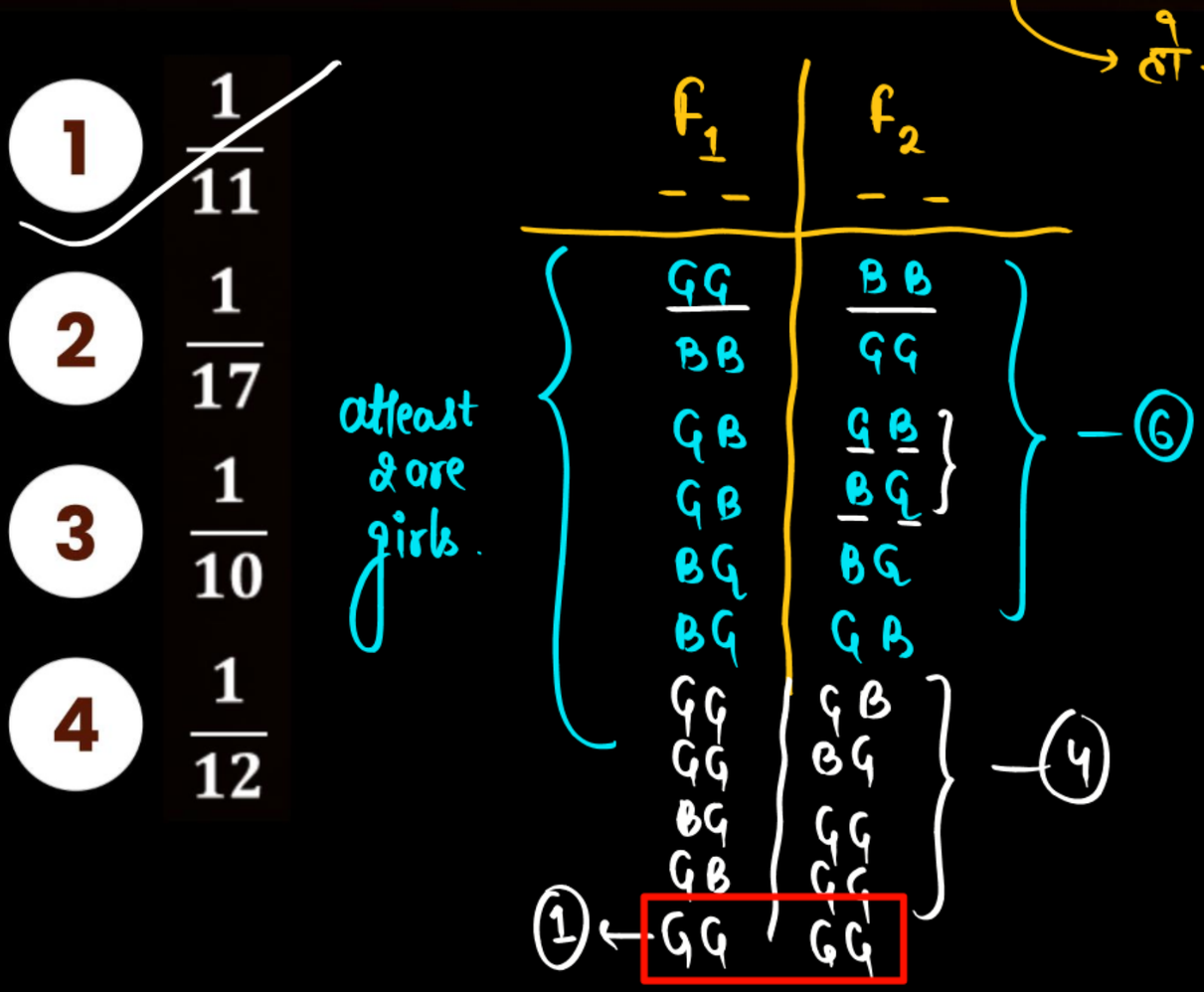
$P(\bar{A} \cap \bar{B}) = t = \frac{4}{10}$.

$\frac{P(\bar{A} \cup \bar{B})}{P(\bar{A} \cap \bar{B})} = \frac{9/10}{4/10} = \frac{9}{4}$

## QUESTION [JEE Mains-2019]

Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is

1. $\frac{1}{11}$
2. $\frac{1}{17}$
3. $\frac{1}{10}$
4. $\frac{1}{12}$



## QUESTION [JEE Mains-2025]

Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains $n$ white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is 29/45, then $n$ is equal to

1. 4
2. 6
3. 3
4. 5

P(White Ball)

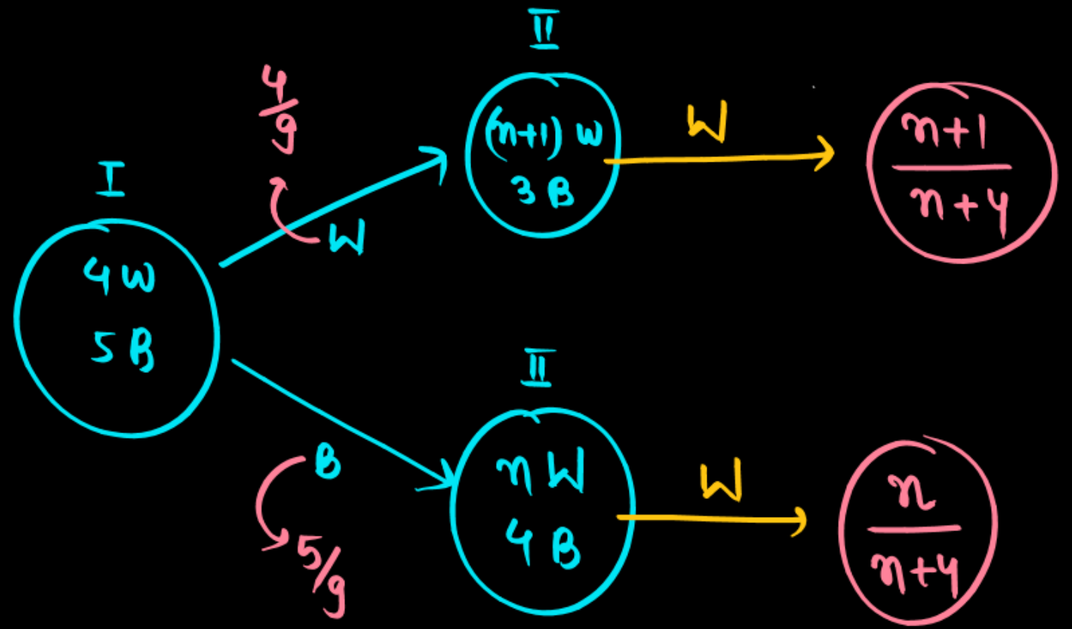
$$\frac{29}{45} = \frac{4}{9}\times\left(\frac{n+1}{n+4}\right) + \frac{5}{9}\times\left(\frac{n}{n+4}\right)$$

$$29(n+4) = 20n + 20 + 25n$$

$$29n + 29\times 4 = 45n + 20$$

$$29\times 4 = 16n$$

$$n = 6$$



## QUESTION [IIT-JEE - 2007]

Repeating

Let $E^C$ denote the complement of an event E. Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$.
Then $P(E^C \cap F^C \mid G)$ equals

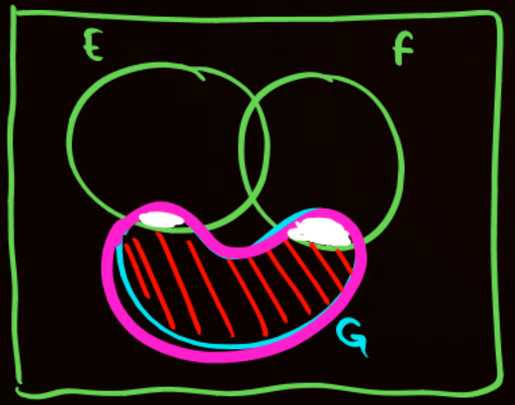
$P(E \cap F) = P(E) \cdot P(F)$
$P(E \cap G) = P(E) \cdot P(G)$
$P(F \cap G) = P(F) \cdot P(G)$

(A) $P(E^c) + P(F^c)$ ✗

(B) $P(E^c) - P(F^c)$ ✗

(C) $P(E^c) - P(F)$ ✓

(D) $P(E) - P(F^c)$ ✗




$$P\left(\frac{\bar{E} \cap \bar{F}}{G}\right) = \frac{P(\bar{E} \cap \bar{F} \cap G)}{P(G)}$$

$$= \frac{P(G) - P(F \cap G) - P(E \cap G)}{P(G)}$$

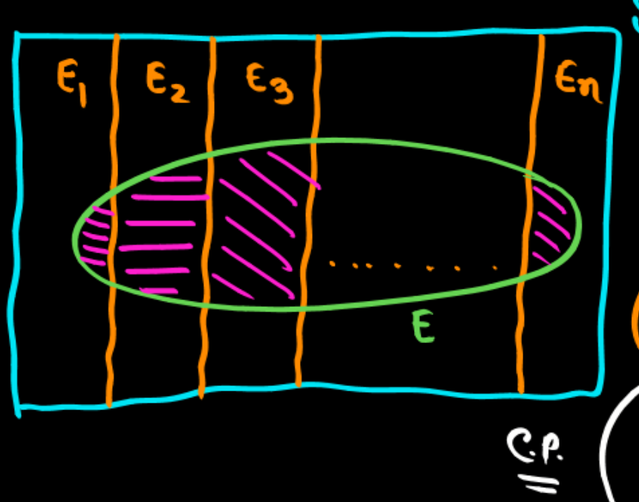
$P(E^c) - P(F)$

$$= \frac{P(G) - P(F) \cdot P(G) - P(E) \cdot P(G)}{P(G)}$$

$P(F^c) - P(E)$

$$= 1 - P(F) - P(E)$$

# TOTAL PROBABILITY THEOREM




Total Prob. of event E = P(E).

$$P(E) = P(E \cap E_1) + P(E \cap E_2) + P(E \cap E_3) + \ldots + P(E \cap E_n)$$

TPT

$$P(E) = P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) + P(E_3) \cdot P\left(\frac{E}{E_3}\right) + \ldots\ldots$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \cdot P\left(\frac{A}{B}\right) = P(A \cap B)$$

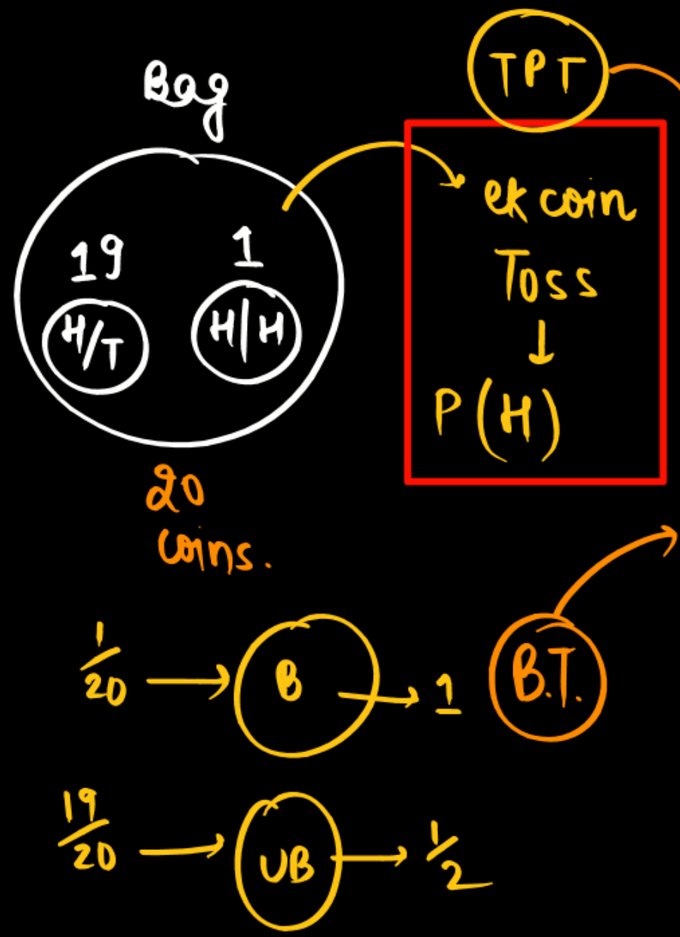
**Baye's Th.** If Event (E) हो चुकी है तो बताओ वो event ($E_2$) के through हुई.

$$P\left(\frac{E_2}{E}\right) = \frac{P(E \cap E_2)}{P(E)} = \frac{P(E_2) \cdot P\left(\frac{E}{E_2}\right)}{\text{TPT.}}$$

## QUESTION [JEE Mains-2025]

A bag contains 19 unbiased coins and one coin with head on both sides. One coin drawn at random is tossed and head turns up. If the probability that the drawn coin was unbiased, is $\frac{m}{n}, \gcd(m,n)=1$, then $n^2-m^2$ is equal to

1. 80
2. 60
3. 72
4. 64




$$P(H)=\frac{19}{20}\times\frac{1}{2}+\frac{1}{20}\times 1$$

(unb. → H, b. → H)

$21^2-19^2=(21-19)(21+19)$

$=2\times 40$

$=80$

$$P\left(\frac{\text{unb.}}{H}\right)=\frac{\frac{19}{20}\times\frac{1}{2}}{\frac{19}{20}\times\frac{1}{2}+\frac{1}{20}\times\frac{2}{2}}=\frac{19}{21}=\frac{m}{n}$$

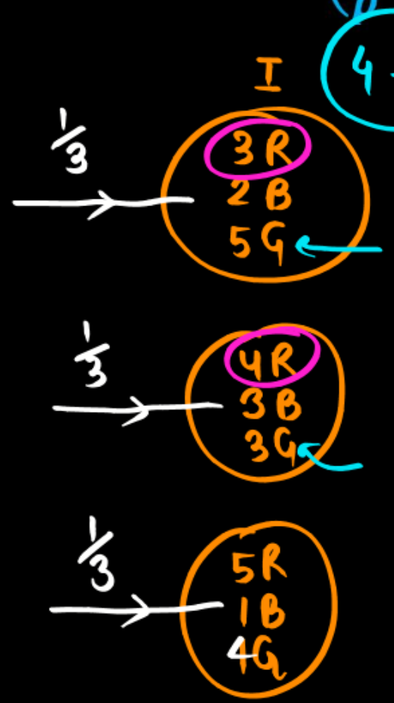
## QUESTION [JEE Mains-2025]

Given three identical bags each containing 10 balls, whose colours are as follows:

| | Red | Blue | Green |
|---|---|---|---|
| Bag I | 3 | 2 | 5 |
| Bag II | 4 | 3 | 3 |
| Bag III | 5 | 1 | 4 |

A person chooses a bag at random and takes out a ball. If the ball is Red, the probability that it is from bag I is $p$ and if the balls is Green, the probability that it is from bag III is $q$, then the value of $\left(\frac{1}{p}+\frac{1}{q}\right)$ is

1. 6
2. 9
3. 7
4. 8




TPT.

$$P(\text{Red ball}) = \frac{1}{3}\times\frac{3}{10}+\frac{1}{3}\times\frac{4}{10}+\frac{1}{3}\times\frac{5}{10}$$

$$P\left(\frac{\text{B-I}}{\text{Red ball}}\right) = \frac{3/30}{\frac{3}{30}+\frac{4}{30}+\frac{5}{30}} = \frac{3}{12} = \frac{1}{4} = p$$

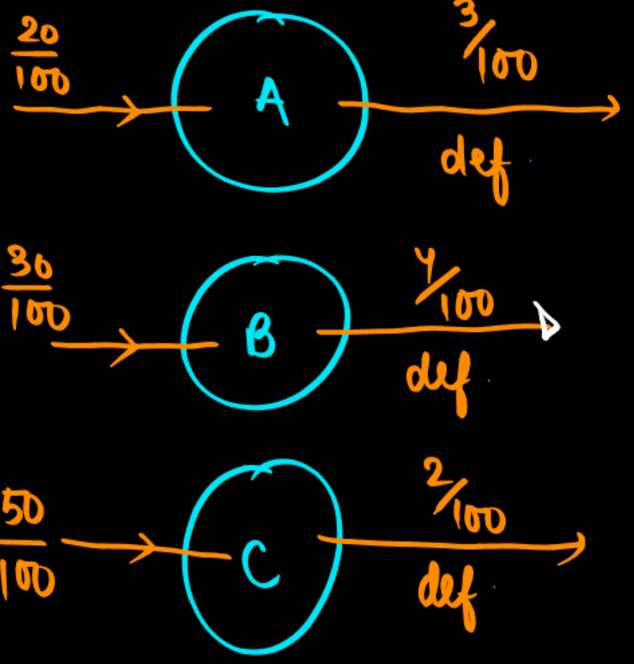
$$P(\text{Green ball}) = \frac{1}{3}\times\frac{5}{10}+\frac{1}{3}\times\frac{3}{10}+\frac{1}{3}\times\frac{4}{10}$$

$$P\left(\frac{\text{B-II}}{\text{Green ball}}\right) = \frac{4/30}{\frac{5}{30}+\frac{3}{30}+\frac{4}{30}} = \frac{4}{12} = \frac{1}{3} = q$$

## QUESTION [JEE Mains-2023]

In a bolt factory, machines $A, B$ and $C$ manufacture respectively 20%, 30% and 50% of the total bolts. Of their output 3, 4 and 2 percent are respectively defective bolts. $A$ bolts is drawn at random from the product. If the bolt drawn is found the defective, then the probability that it is manufactured by the machine $C$ is

1. $\frac{5}{14}$
2. $\frac{9}{28}$
3. $\frac{3}{7}$
4. $\frac{2}{7}$




TPT

$$P(\text{def bolt}) = \frac{20}{100}\times\frac{3}{100} + \frac{30}{100}\times\frac{4}{100} + \frac{50}{100}\times\frac{2}{100}$$

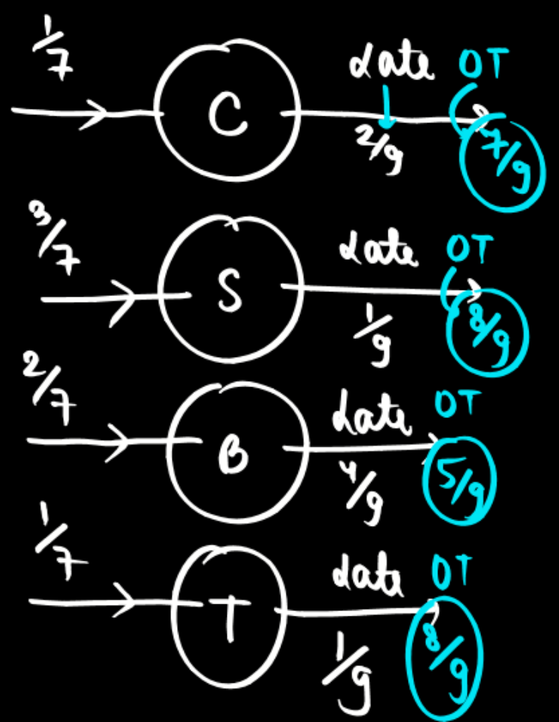
BT

$$P\left(\frac{M-C}{\text{def bolt}}\right) = \frac{\frac{100}{10^4}}{\frac{60}{10^4}+\frac{120}{10^4}+\frac{100}{10^4}} = \frac{100}{280} = \frac{5}{14}$$

## QUESTION [IIT-JEE-2005]

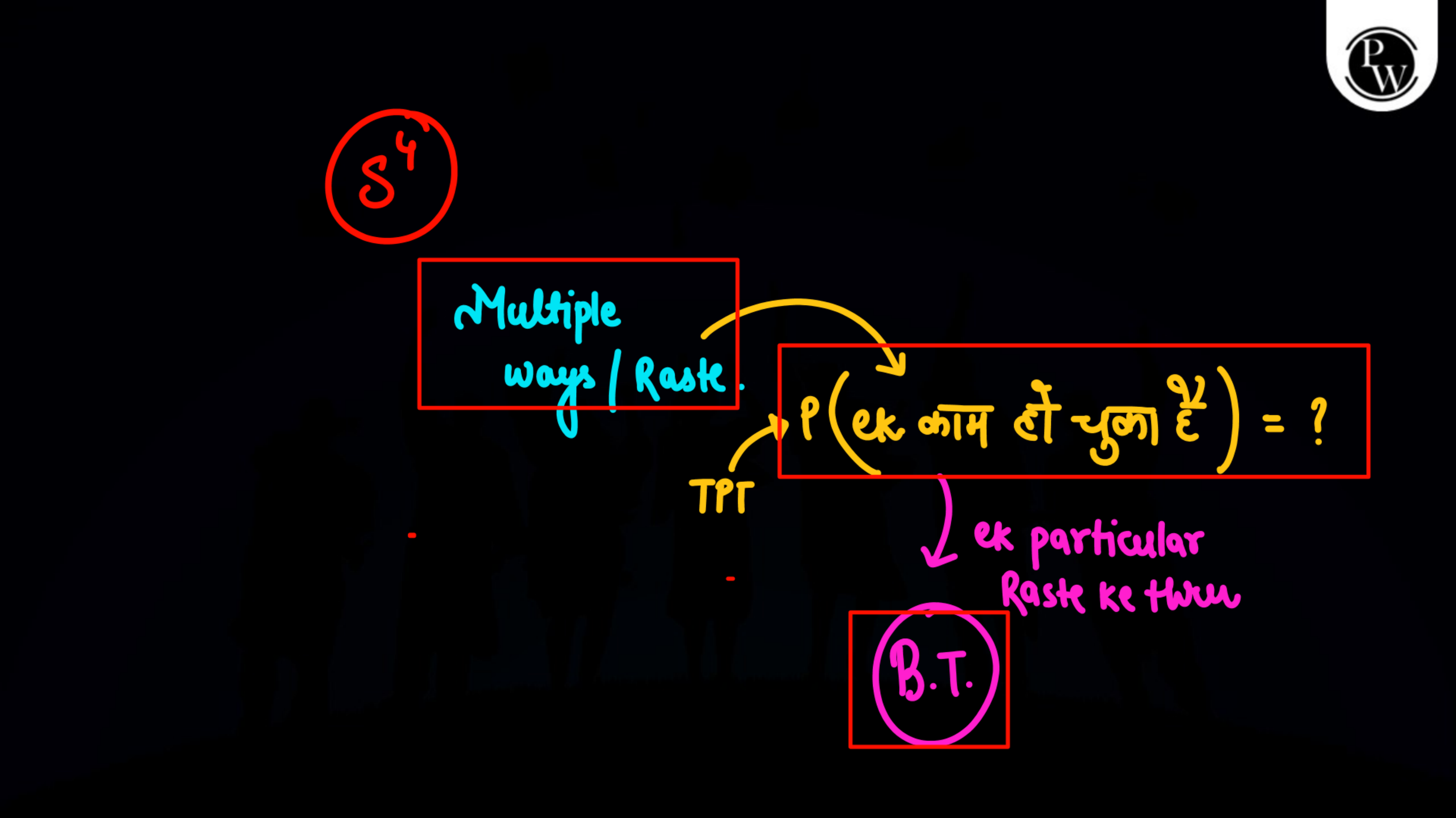
A person goes to office either by car, scooter, bus or train probability of which being $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$, respectively. Probability that he reaches offices late, if he takes car, scooter, bus or train is $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$, respectively. Given that he reached office in time, then what is the probability that he travelled by a car?

(हो चुका है → TPT)




$$P(\text{on Time}) = \frac{1}{7}\times\frac{7}{9} + \frac{3}{7}\times\frac{8}{9} + \frac{2}{7}\times\frac{5}{9} + \frac{1}{7}\times\frac{8}{9}$$

$$P\left(\frac{\text{Car}}{\text{on Time}}\right) = \frac{7/63}{\frac{7}{63} + \frac{24}{63} + \frac{10}{63} + \frac{8}{63}} = \frac{7}{7+42} = \frac{1}{1+6} = \frac{1}{7}$$

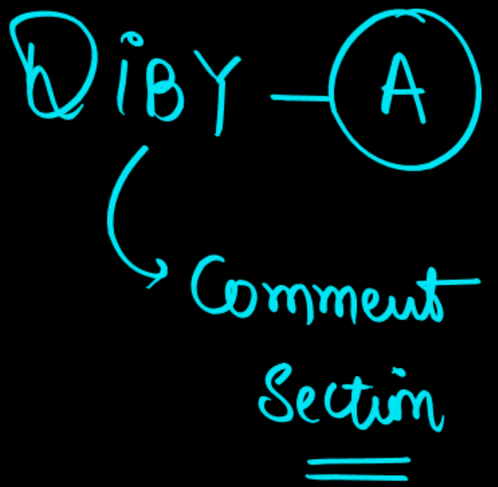

S4
Multiple ways / Raste
P(ek काम हो चुका है) = ?
TPT
ek particular Raste ke thru
B.T.

## QUESTION [JEE Mains-2021]

NCERT

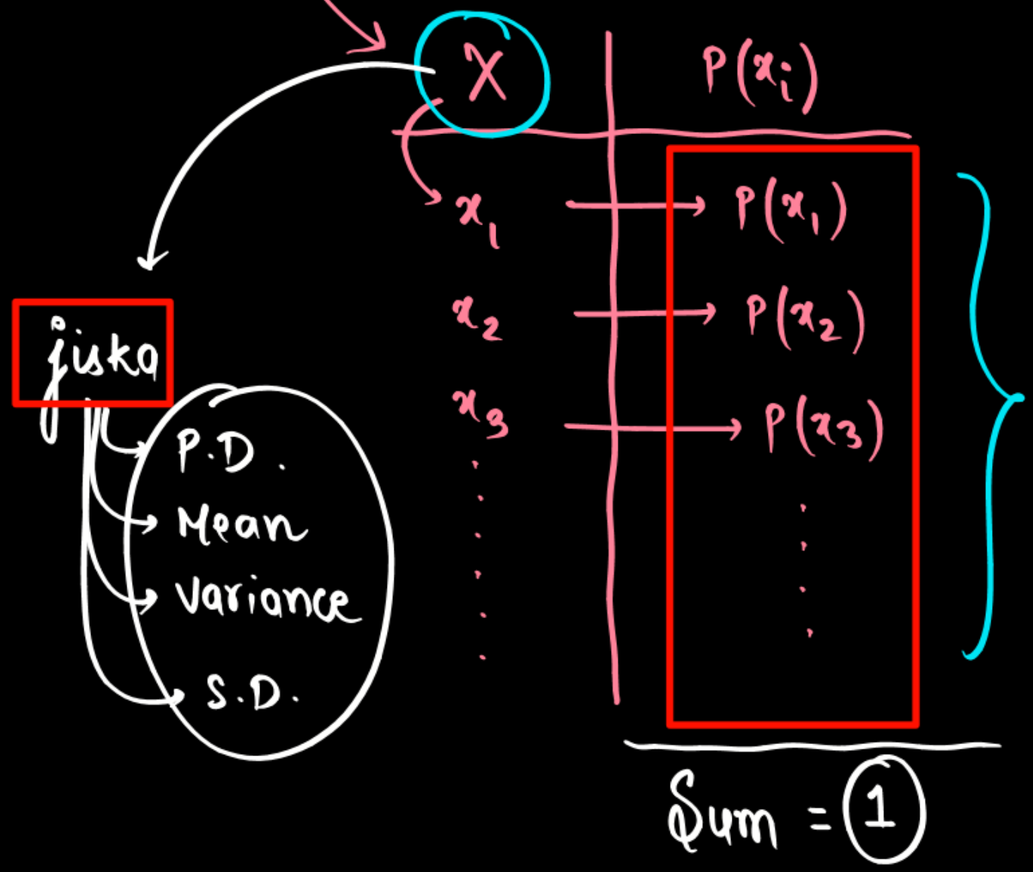
In a group of 400 people, 160 are smokers and nonvegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are 35%, 20% and 10% respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is

1. $\frac{8}{45}$
2. $\frac{7}{45}$
3. $\frac{28}{45}$
4. $\frac{14}{45}$



# RANDOM VARIABLE

Viral + Trending



## QUESTION [JEE Mains-2020]

**A random variable $X$ has the following probability distribution :**

| $X$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X)$ | $K^2$ | $2K$ | $K$ | $2K$ | $5K^2$ |

**Then $P(X > 2)$ is equal to**

1. $\frac{7}{12}$
2. $\frac{23}{36}$
3. $\frac{1}{36}$
4. $\frac{1}{6}$

| $X$ | $P(X)$ |
|---|---|
| 1 | $k^2$ |
| 2 | $2k$ |
| 3 | $k$ |
| 4 | $2k$ |
| 5 | $5k^2$ |

Sum = 1.

$6k^2 + 5k = 1$

$6k^2 + 5k - 1 = 0$

$k = \frac{1}{6}$ ✓

$k = -1$ ✗

$P(X > 2) = P(X=3) + P(X=4) + P(X=5)$

$= k + 2k + 5k^2$

$= \frac{1}{6} + \frac{2}{6} + \frac{5}{36} = \frac{6+12+5}{36} = \frac{23}{36}$

## QUESTION [JEE Mains-2020]

**A random variable $X$ has the following probability distribution :**

| $X$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X)$ | $K^2$ | $2K$ | $K$ | $2K$ | $5K^2$ |

**Then $P(X > 2)$ is equal to**

1. $\frac{7}{12}$
2. $\frac{23}{36}$
3. $\frac{1}{36}$
4. $\frac{1}{6}$

$\#\ P(X \geq 2) = 2k + k + 2k + 5k^2$

OR

$= 1 - k^2 = 1 - \frac{1}{36} = \frac{35}{36}$.

$\bar{X}$ = Mean $= 1 \times k^2 + 2 \times 2k + 3 \times k + 4 \times 2k + 5 \times 5k^2$

Variance $(\sigma_X^2) = \left(1 \times k^2 + 4 \times 2k + 9 \times k + 16 \times 2k + 25 \times 5k^2\right) - (\bar{X})^2$

## QUESTION [JEE Mains-2025]

If the probability that the random variable $X$ takes the value $x$ is given by $P(X = x) = k(x + 1)3^{-x}, x = 1, 2, 3, \ldots$, where $k$ is a constant, then $P(X \geq 3)$ is equal to

1. $\frac{7}{27}$
2. $\frac{4}{9}$
3. $\frac{8}{27}$
4. $\frac{1}{9}$

| $X = x$ | $P(X = x)$ |
|---|---|
| $x = 1$ | $\frac{2k}{3}$ |
| $x = 2$ | $\frac{3k}{3^2}$ |
| $x = 3$ | $\frac{4k}{3^3}$ |
| ⋮ | ⋮ |
| Sum = 1 | |

AGP

$$1 = \frac{2k}{3} + \frac{3k}{3^2} + \frac{4k}{3^3} + \frac{5k}{3^4} + \ldots\ldots\infty$$

$$\left(\frac{1}{3}\right)\times \quad \frac{1}{3} = 0 + \frac{2k}{3^2} + \frac{3k}{3^3} + \frac{4k}{3^4} + \ldots\ldots\infty$$

$$\ominus$$

$$\frac{2}{3} = \frac{2k}{3} + \left(\frac{k}{3^2} + \frac{k}{3^3} + \frac{k}{3^4} + \ldots\ldots\infty\right)$$

$$\frac{2}{3} = \frac{2k}{3} + \frac{k}{3^2} \times \frac{3}{2}$$

$$\frac{2}{3} = k\left(\frac{2}{3} + \frac{1}{6}\right)$$

$$\frac{5k}{6} = \frac{2}{3}$$

$$k = \frac{4}{5}$$

## QUESTION [JEE Mains-2025]

If the probability that the random variable $X$ takes the value $x$ is given by $P(X = x) = k(x + 1)3^{-x}, x = 1, 2, 3, \ldots$, where $k$ is a constant, then $P(X \geq 3)$ is equal to

| $X = x$ | $P(X = x)$ |
|---|---|
| $x = 1$ | $\frac{2k}{3}$ |
| $x = 2$ | $\frac{3k}{3^2}$ |
| $x = 3$ | $\frac{4k}{3^3}$ |
| $\vdots$ | $\vdots$ |
| | $\vdots$ |
| Sum = 1 | |

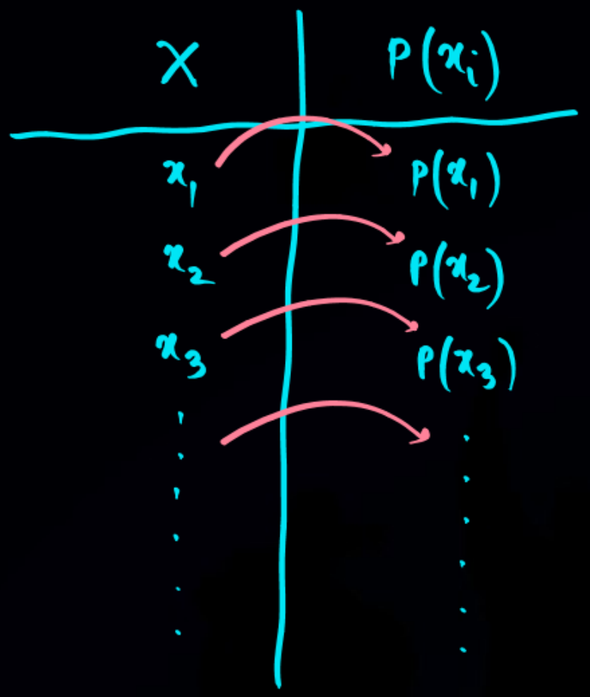
$$P(X \geq 3) = \frac{4k}{3^3} + \frac{5k}{3^4} + \frac{6k}{3^5} + \ldots \infty$$

$$= 1 - \left(\frac{2k}{3} + \frac{k}{3}\right)$$

$$= 1 - k$$

$$= 1 - \frac{4}{5} = \frac{1}{5}$$

$$k = \frac{4}{5}$$




Mean ($\bar{X}$) = expected value = avg.

$= x_1 P(x_1) + x_2 P(x_2) + x_3 P(x_3) + \ldots$

$$\mu = \bar{x} = \sum x_i P(x_i)$$

Variance ($\sigma^2$ or $\sigma_x^2$) $= \left(\sum x_i^2 P(x_i)\right) - (\text{Mean})^2$

$\sigma$ = S.D.

# MEAN, VARIANCE & S.D.

**Mean :**

**Denote as :** $\mu$ **or** $\bar{X}$ **or E(X) = Expected value of X.**

$$\mu = \bar{X} = E(X) = \frac{\sum p_i . X_i}{\sum p_i} = \sum p_i . X_i$$

**Variance ($\sigma_X^2$):**

**Variance :** $V(X) = \sigma_X^2 = \sum p_i (X_i - \mu)^2$

$$= \sigma_X^2 = \sum p_i (X_i - \mu)^2 = \sum p_i X_i^2 - \left(\sum p_i X_i\right)^2$$

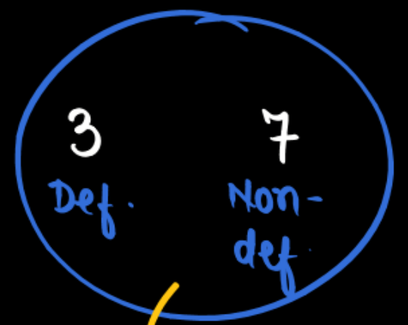
Mean.

**Standard Deviation** $(\sigma_x) = \sqrt{\text{Variance}}$

## QUESTION [JEE Mains-2025]

A box contains 10 pens of which 3 are defective. A sample of 2 pens is drawn at random and let $X$ denote the number of defective pens. Then the variance of $X$ is

1. $\frac{11}{15}$
2. $\frac{28}{75}$
3. $\frac{2}{15}$
4. $\frac{3}{5}$




2 pens select.

| $X$ = no. of def. Pens | $P(X)$ |
|---|---|
| 0 | $\frac{{}^7C_2}{{}^{10}C_2} = \frac{21}{45}$ |
| 1 | $\frac{{}^3C_1 \cdot {}^7C_1}{{}^{10}C_2} = \frac{21}{45}$ |
| 2 | $\frac{{}^3C_2}{{}^{10}C_2} = \frac{3}{45}$ |

$$\bar{X} = 0\times\frac{21}{45} + 1\times\frac{21}{45} + 2\times\frac{3}{45}$$

$$= \frac{27}{45} = \frac{3}{5}$$

Variance $(\sigma_x^2)$

$$= \left(0\times\frac{21}{45} + 1\times\frac{21}{45} + 4\times\frac{3}{45}\right) - \left(\frac{3}{5}\right)^2$$

$$= \frac{\frac{33}{45}}{15} - \frac{9}{25} = \frac{55-27}{75} = \frac{28}{75}$$

**QUESTION [JEE Mains-2025]**

Tail follows a head ≡ Head के बाद Tail.

A coin is tossed three times. Let $X$ denote the number of times a tail follows a head. If $\mu$ and $\sigma^2$ denote the mean and variance of $X$, then the value of $64(\mu + \sigma^2)$ is

1. 64
2. 48
3. 51
4. 32

X

HHH → 0
HHT → 1
HTH → 1
THH → 0
HTT → 1
THT → 1
TTH → 0
TTT → 0

$\frac{1}{8}$ ←

| X | P(X) |
|---|---|
| 0 | $4 \times \frac{1}{8}$ |
| 1 | $4 \times \frac{1}{8}$ |

| X | P(X) |
|---|---|
| 0 | $\frac{1}{2}$ |
| 1 | $\frac{1}{2}$ |

$\mu = \bar{X} = \frac{1}{2}$

$\sigma_X^2 = 0 + \frac{1}{2} - \left(\frac{1}{2}\right)^2$

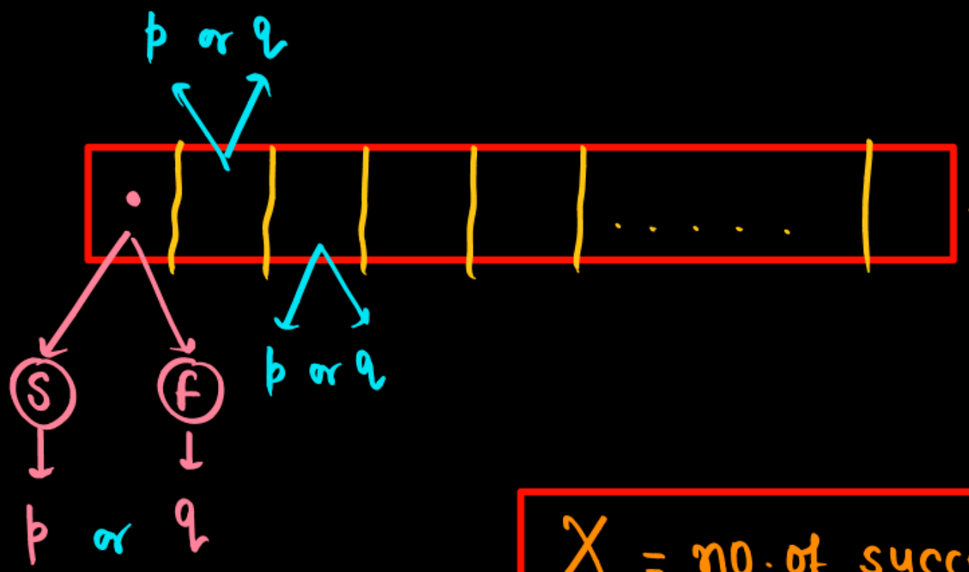
$\sigma_X^2 = \frac{1}{4}$

$64\left(\frac{1}{2} + \frac{1}{4}\right) = 32 + 16$

$= 48$

# BERNOULLI'S TRIAL

## Binomial Probability Distribution (B.P.D.)




→ $n$ Trials / $n$-Times exp. is performed.

→ finite.

$p + q = 1$.

$X$ = no. of successes.

| $X$ | $P(X)$ |
|---|---|
| 0 | ${}^nC_0\, q^n$ |
| 1 | ${}^nC_1 \times p^1 \times q^{n-1}$ |
| 2 | ${}^nC_2 \times p^2 \times q^{n-2}$ |
| 3 | ${}^nC_3 \times p^3 \times q^{n-3}$ |
| ⋮ | ⋮ |
| $n$ | ${}^nC_n \times p^n$ |

Note: $B(n, p)$ — $n$: no. of Trials; $p$: prob. of success in each trial; B.T.

$P(X = r)$ = Prob. of $r$ – successes.

$$P(X = r) = {}^nC_r\, p^r\, q^{n-r}$$

$$\bar{X} = np \quad , \quad \sigma_x^2 = npq$$

# MEAN, VARIANCE & S.D.

Mean $(\overline{X})$ = $np$

Variance $(\sigma_x^2)$ = $npq$

Standard Deviation $(\sigma_x)$ = $\sqrt{npq}$

Ques :-

$B\left(10, \frac{1}{4}\right)$ — $n$, $p$

Binom. prop. dist.

Mean & Variance of Binomial Variate

$n = 10$

$p = \frac{1}{4}$

$q = \frac{3}{4}$

$\bar{X} = 10 \times \frac{1}{4} = \frac{5}{2}$

$\sigma_X^2 = \cancel{10}^{5} \times \frac{1}{\cancel{4}_{2}} \times \frac{3}{4} = \frac{15}{8}$

$P(X=4) = {}^{10}C_4 \times \left(\frac{1}{4}\right)^4 \times \left(\frac{3}{4}\right)^6$

4 successes.

Prob of atleast 1 success $= P(X \geq 1)$

$= 1 - P(X=0)$

$= 1 - \left(\frac{3}{4}\right)^{10}$

Prob of atleast 2 success $= P(X \geq 2)$

$= 1 - P(X=0) - P(X=1)$

## QUESTION [JEE Mains-2023]

In a binomial distribution $B(n, p)$, the sum and product of the mean and variance are 5 and 6 respectively, then find $6(n+p-q)$ is equal to

1. 51
2. 52
3. 53
4. 50

$\bar{X} = np$

$\sigma_x^2 = npq$

$np + npq = 5$.

$(np)(npq) = 6$

$p + q = 1$.

SBS

$np(1+q) = 5$

$np^2q = 6$

$np^2(1+q)^2 = 25$

$\frac{q}{(q+1)^2} = \frac{6}{25}$

$25q = 6q^2 + 6 + 12q$

$6q^2 - 13q + 6 = 0$

$6q^2 - 9q - 4q + 6 = 0$

$3q(2q-3) - 2(2q-3) = 0$

$q = \frac{2}{3}, \frac{3}{2}$ (✓, ✗)

$p = \frac{1}{3}$

$n \times \frac{1}{3} \times \frac{5}{3} = 5$

$n = 9$

$6\left(9 + \frac{1}{3} - \frac{2}{3}\right) = 6\left(9 - \frac{1}{3}\right)$

$= 54 - 2$

$= 52$

Ans.

## QUESTION [JEE Mains-2022]

A random variable $X$ follows the Binomial distribution $B(33, p)$ such that $3P(X=0) = P(X=1)$, then the value of $\frac{P(X=15)}{P(X=18)} - \frac{P(X=16)}{P(X=17)}$ is equal to

1. 1320
2. 1088
3. $\frac{120}{1331}$
4. $\frac{1088}{1089}$

$n = 33$

$3 \times {}^{33}C_0 (q)^{33} = {}^{33}C_1 \times p^1 \cdot (q)^{32}$

$3q = 33p$

$q = 11p$

$p + q = 1$

$12p = 1$

$p = \frac{1}{12}, \quad q = \frac{11}{12} \Rightarrow \frac{q}{p} = 11$

$\frac{{}^{33}C_{15} p^{15} q^{18}}{{}^{33}C_{18} p^{18} q^{15}} - \frac{{}^{33}C_{16} p^{16} q^{17}}{{}^{33}C_{17} p^{17} q^{16}}$

$\left(\frac{q}{p}\right)^3 - \frac{q}{p} = 11^3 - 11$

$= 11 \times (11^2 - 1)$

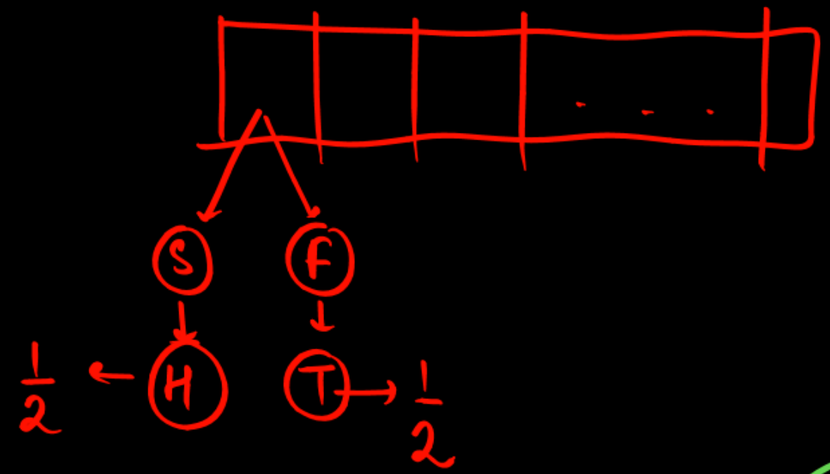
$11 \times 12 \times 10$

$= 1320$

## QUESTION [JEE Mains-2021]

A fair coin is tossed $n$-times such that the probability of getting at least one head is at least 0.9. Then the ~~minimum~~ least value of $n$ is

9 Ans.




$$P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - \left(\frac{1}{2}\right)^n \geq 0.9$$

$$1 - \frac{1}{2^n} \geq \frac{9}{10}$$

$$2^n \geq 10$$

$$\frac{1}{10} \geq \frac{1}{2^n}$$

n=1 ✗ n=2 ✗ n=3 ✗ n=4 ✓ n=5 ✓ n=6 ✓ ...

least (n)

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# Thank You